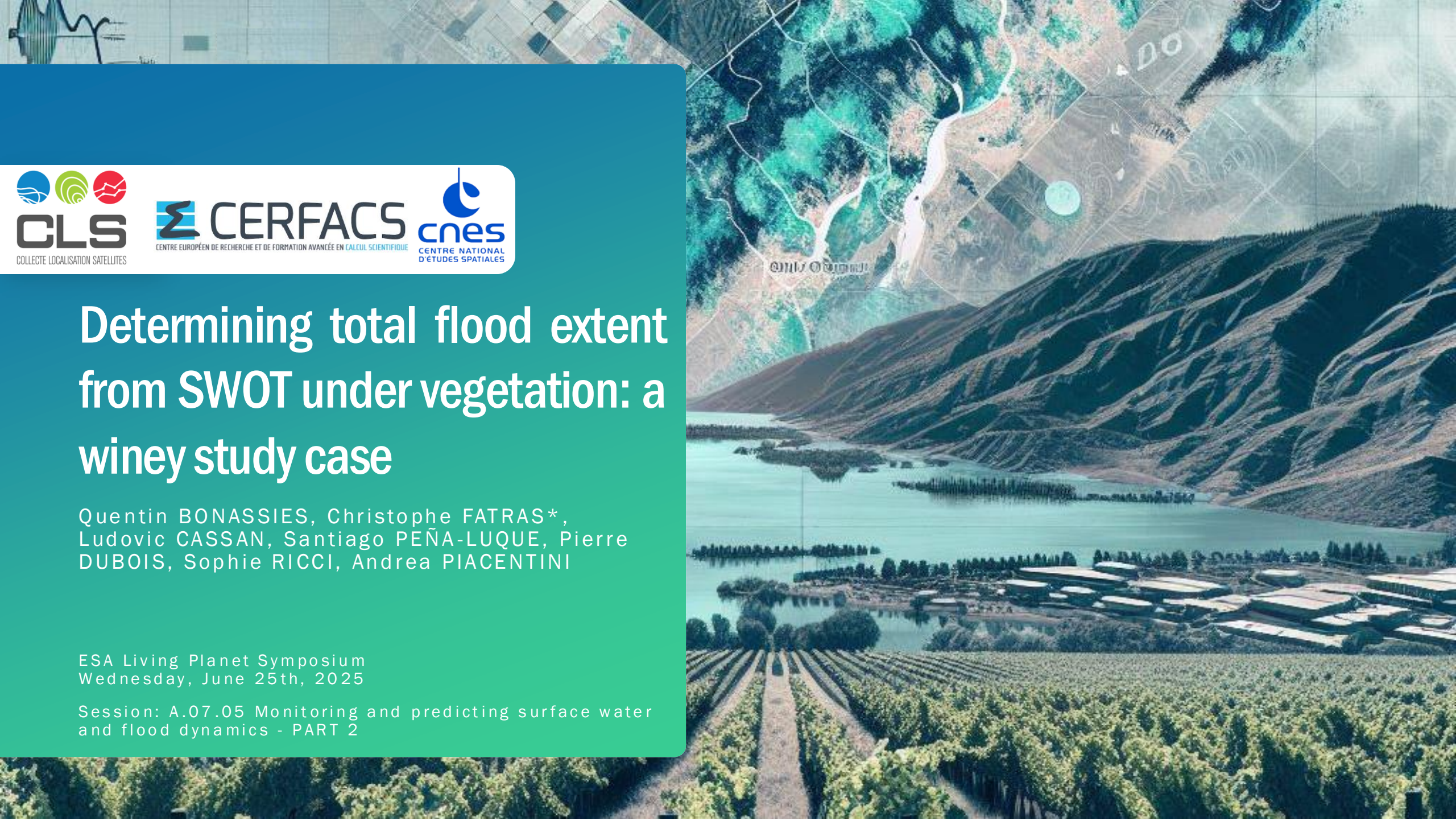


Determining total flood extent from SWOT under vegetation: a winey study case

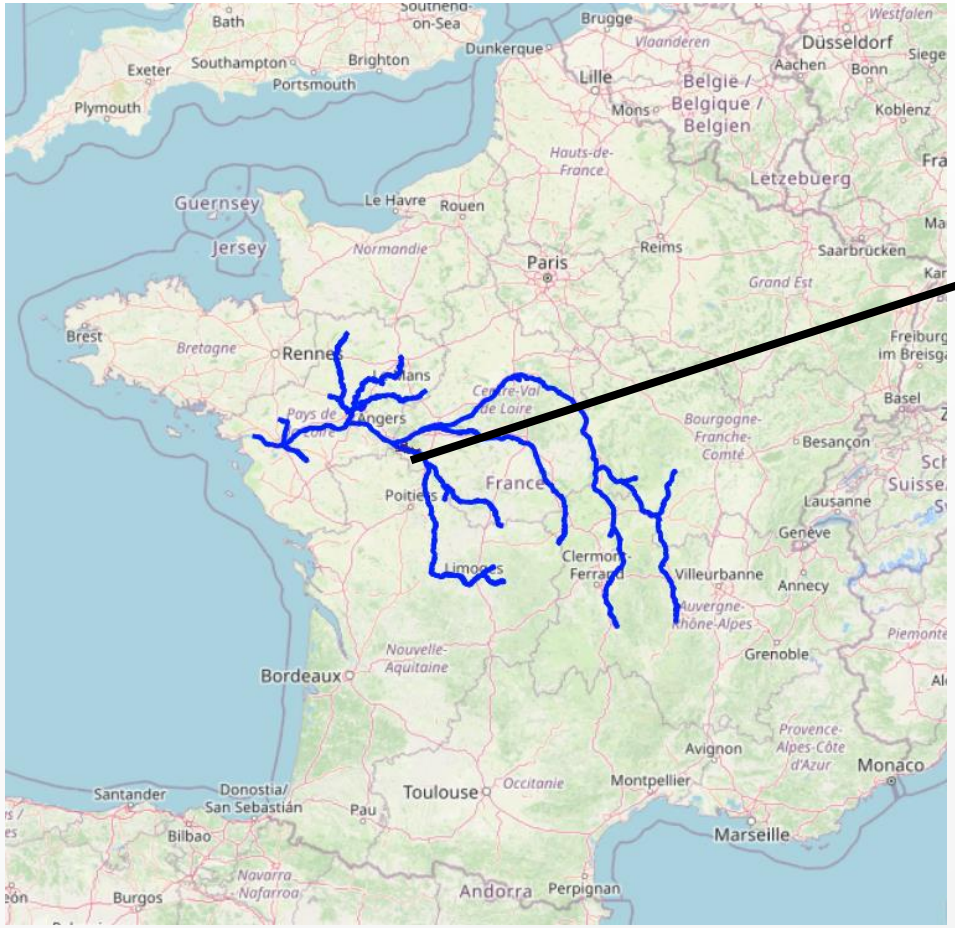
Quentin BONASSIES, Christophe FATRAS*, Ludovic CASSAN, Santiago PEÑA-LUQUE, Pierre DUBOIS, Sophie RICCI, Andrea PIACENTINI

ESA Living Planet Symposium
Wednesday, June 25th, 2025

Session: A.07.05 Monitoring and predicting surface water and flood dynamics - PART 2

Study genesis

Wine production area in the Loire valley, close to Tours: Chinon!



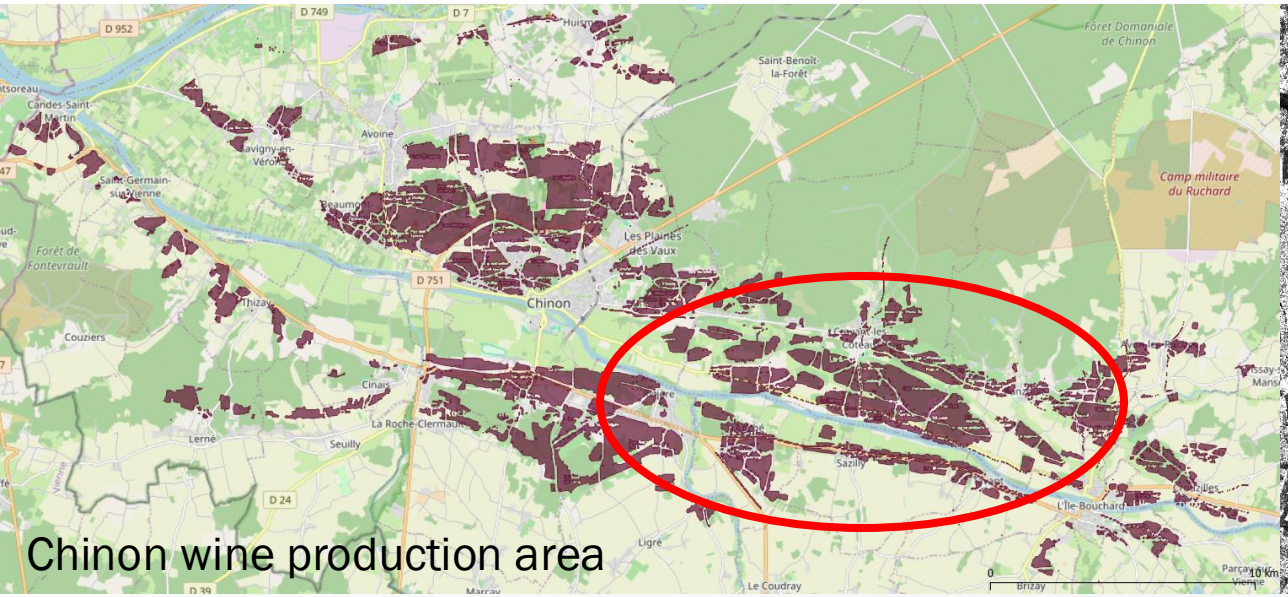




But sometimes, the area is flooded…

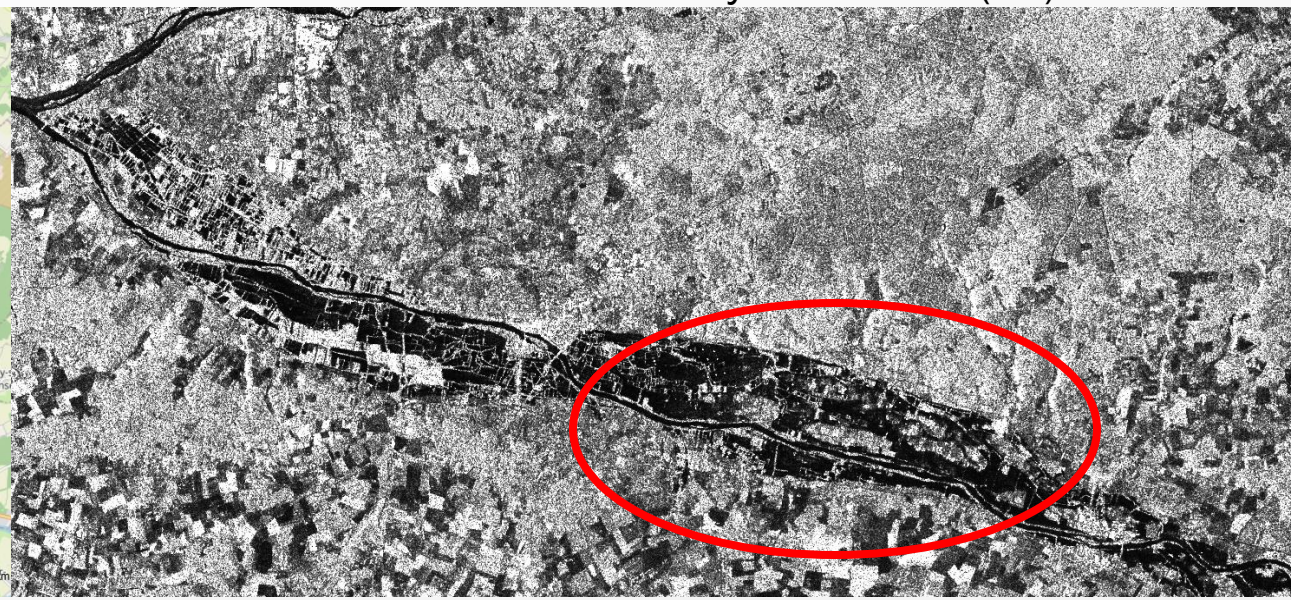
Study genesis

Vienne river flood… on March 31st, 2024



Chinon wine production area

As observed by Sentinel-1 (VV)



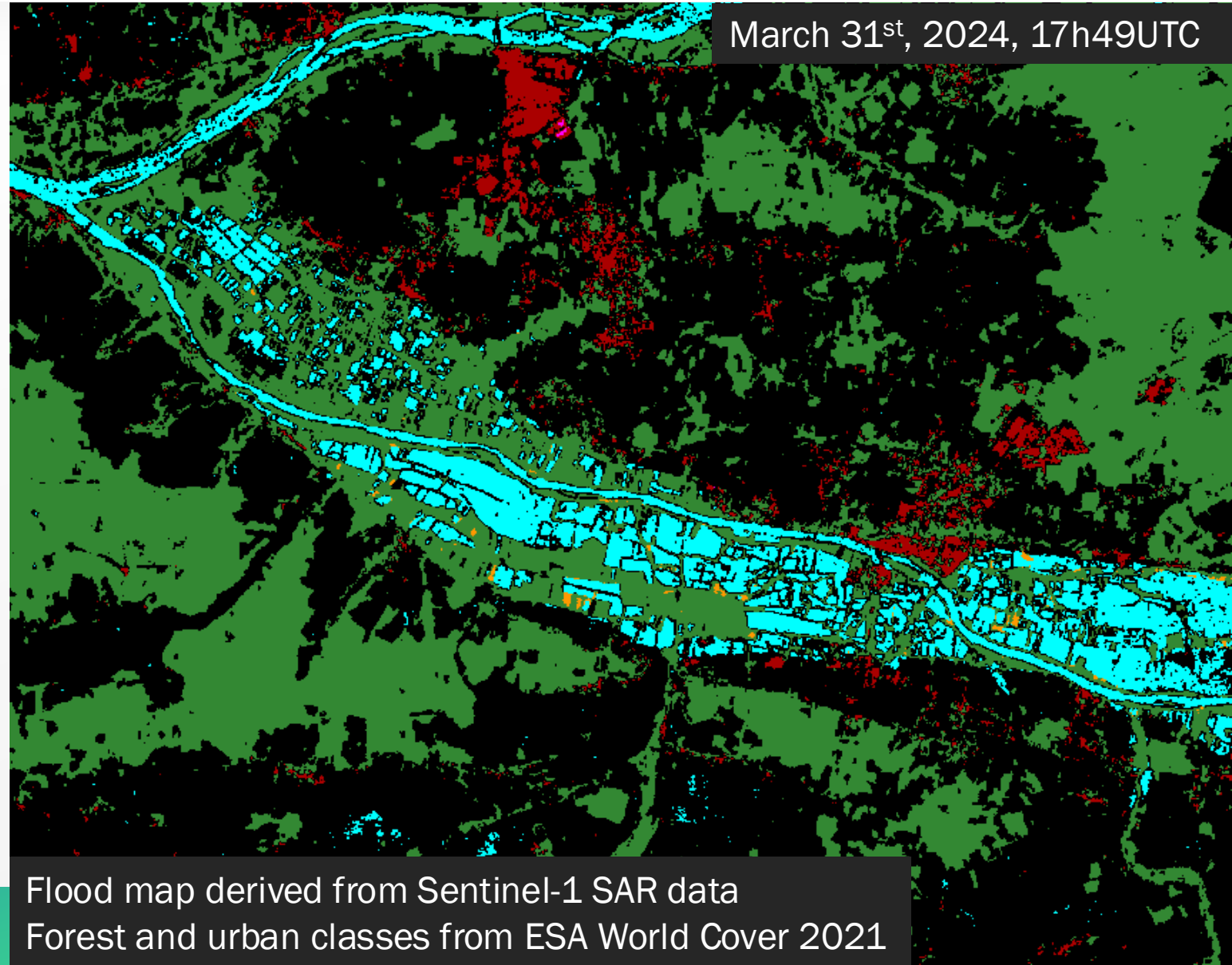
17h49 UTC

Vineyards clearly affected by flooding, but the rest…..

What is the issue with traditional flood mapping?

- Flooded forest cannot be observed directly
- Flooded urban areas are not « easily » detected
- Combination with DEM powerful, but relies strongly on the quality of all ancillary data

By curiosity, is there any close SWOT data?



Flood map derived from Sentinel-1 SAR data
Forest and urban classes from ESA World Cover 2021

Study genesis

Yes!

Vienne river flood… on March 31st, 2024





17h49 UTC → Only 2h33 apart! ← 15h16 UTC

What can SWOT bring to the study of flooded areas compared to side-looking SAR and optical imagery?

SWOT – An interferometric close nadir SAR

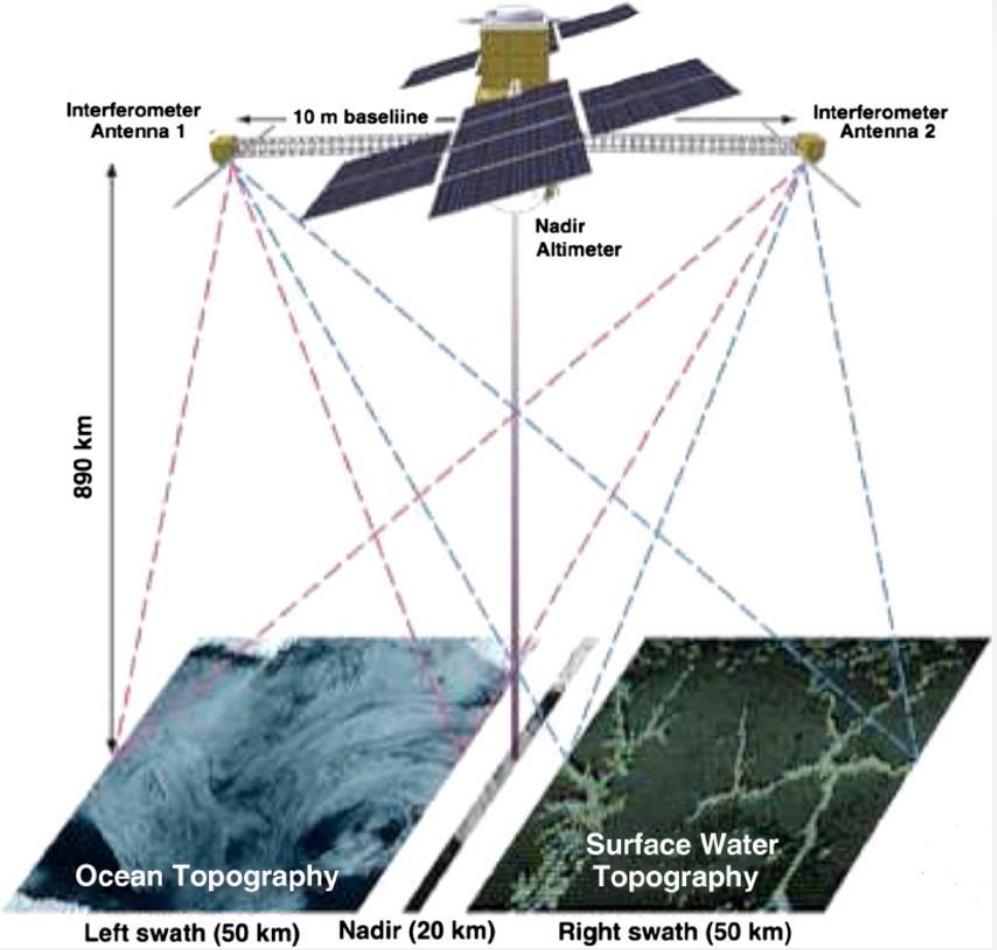
New 2D SAR close to nadir altimeter, launched in December 2022

Product used:

The L2_HR_**PIXC** (Pixel Cloud)

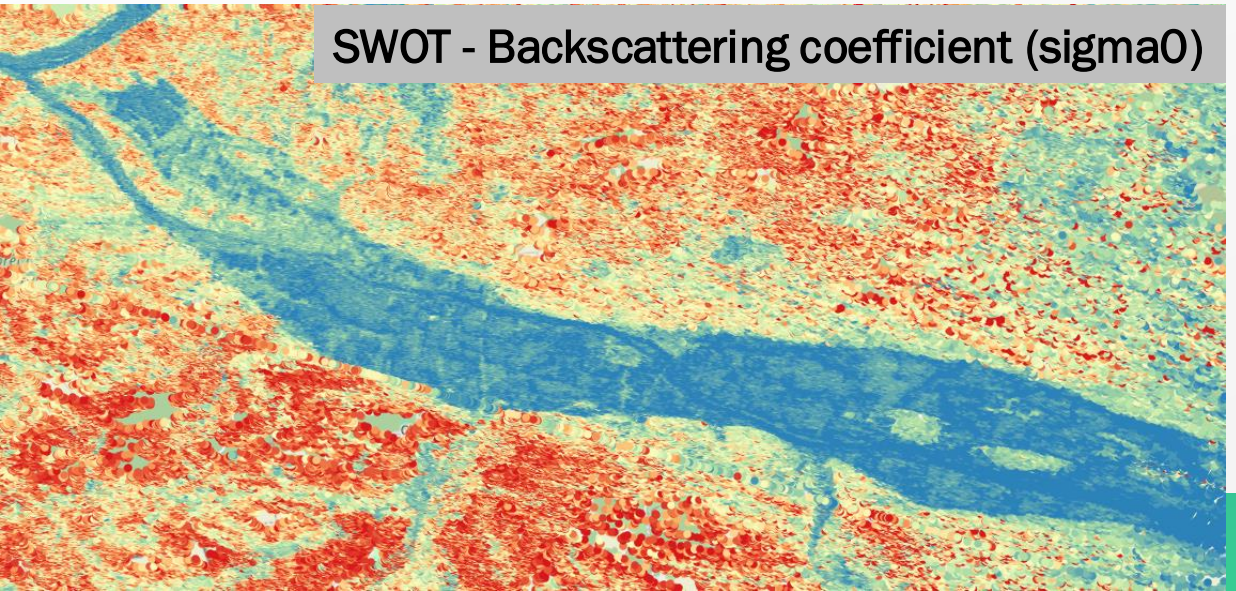
- Many, many variables, among them:
  - "Native" classification of PIXC (water/land, etc.)
  - Backscattering coefficient (sigma0)
  - Coherent power (P_{coh})
  - Interferometric coherence (γ_{tot})
  - SNR
  - … so many information!

$$\gamma_{total} = \frac{\langle s_1^* s_2 \rangle}{\sqrt{\langle |s_1|^2 \rangle \langle |s_2|^2 \rangle}}$$

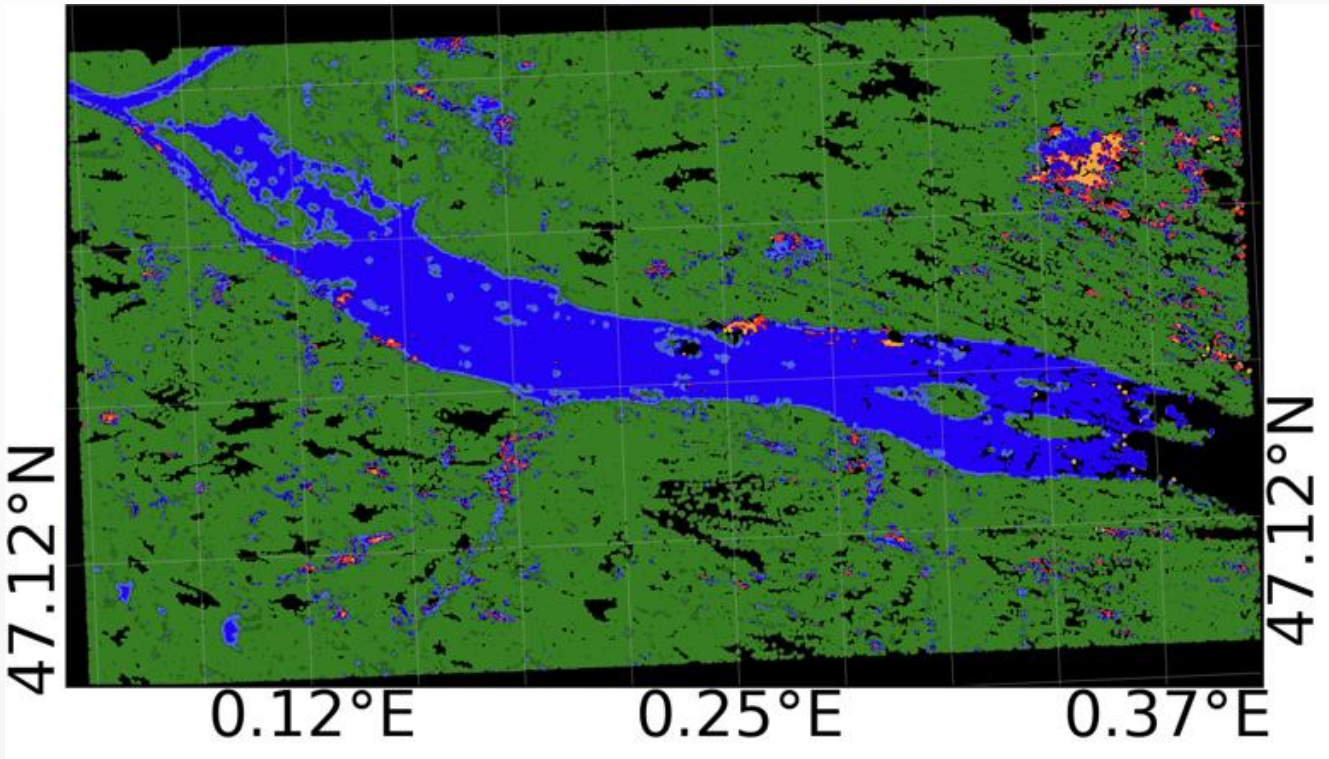


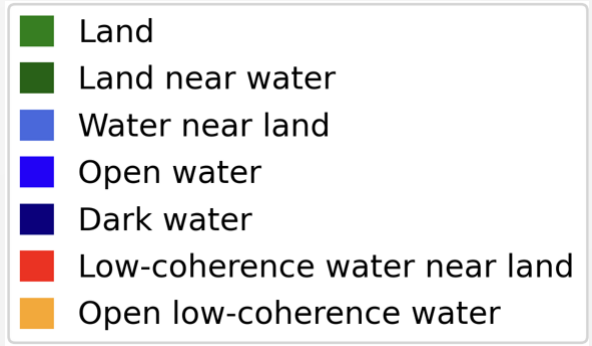
SWOT - Outputs on the Chinon case



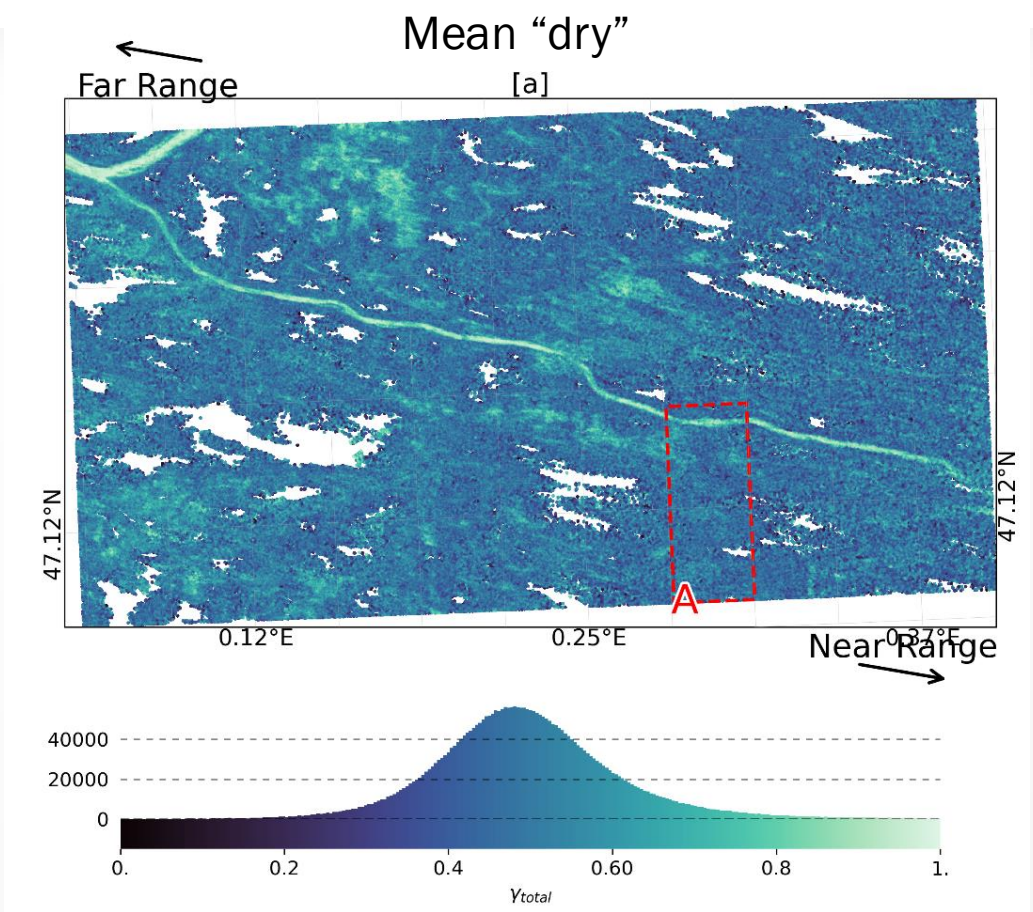


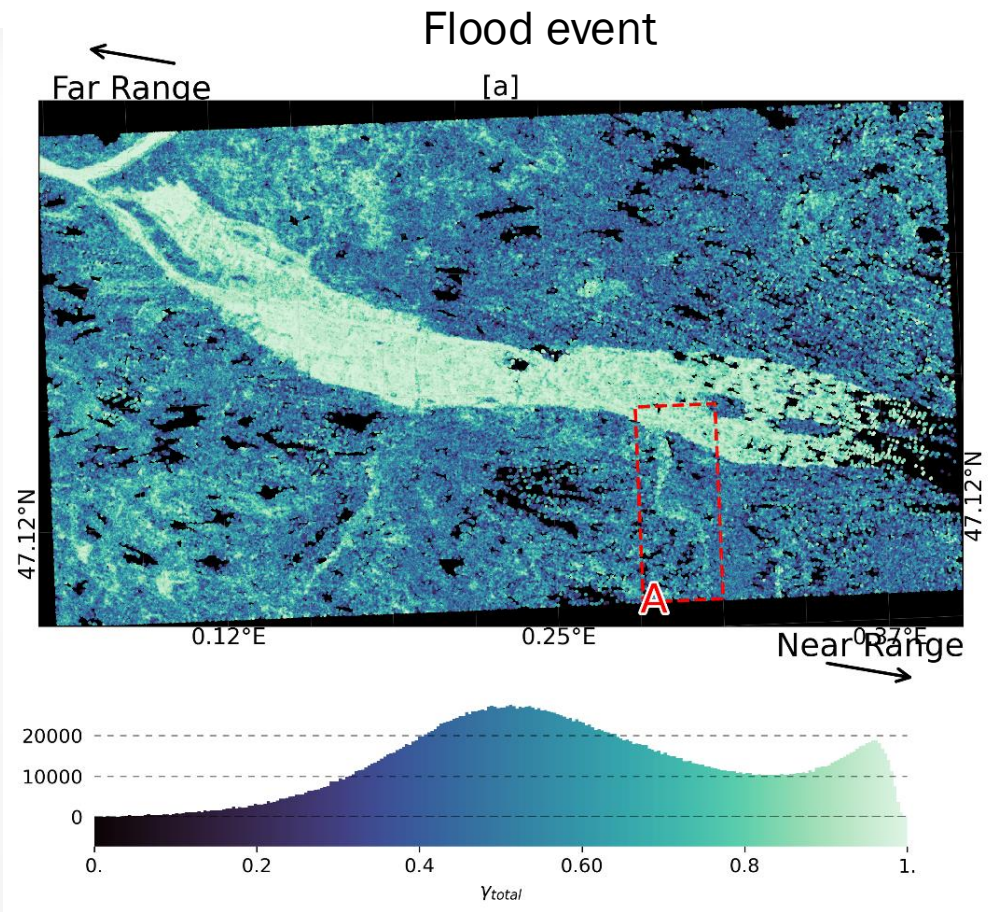
PIXC “native” classification

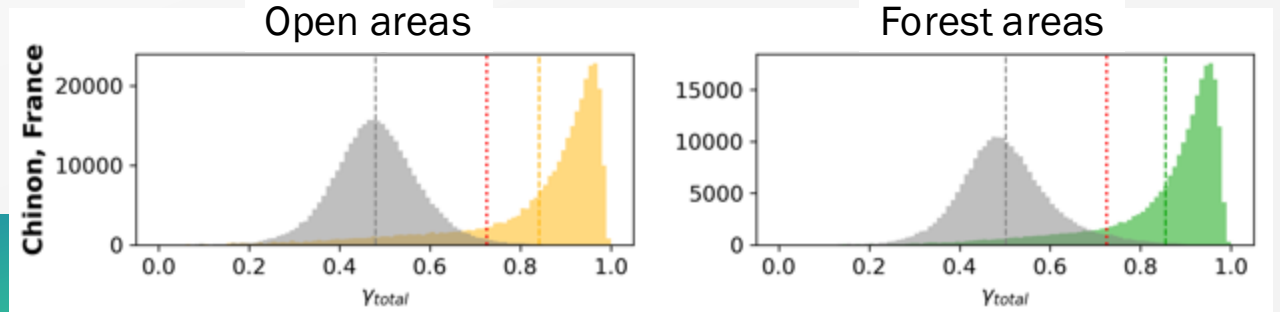




SWOT – Interferometric coherence

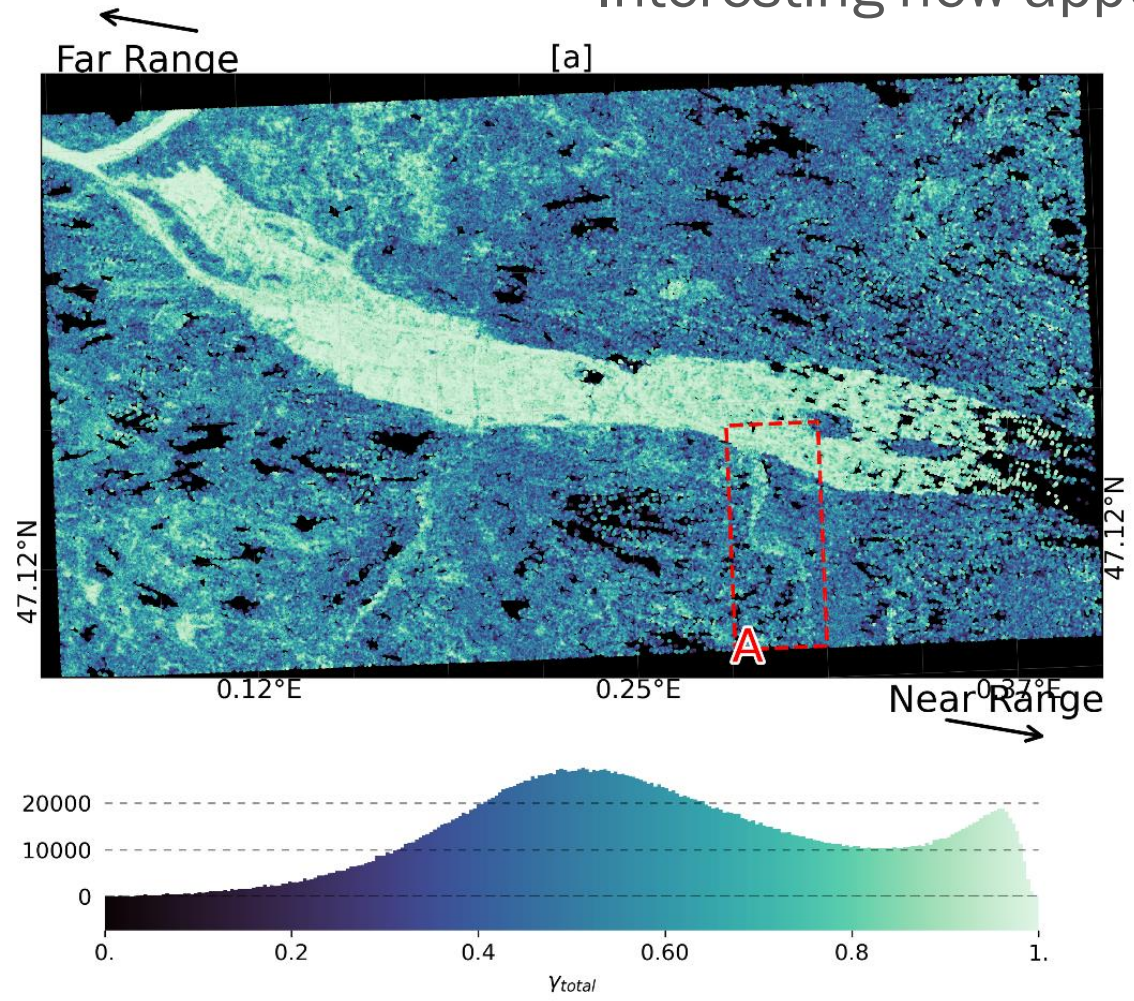


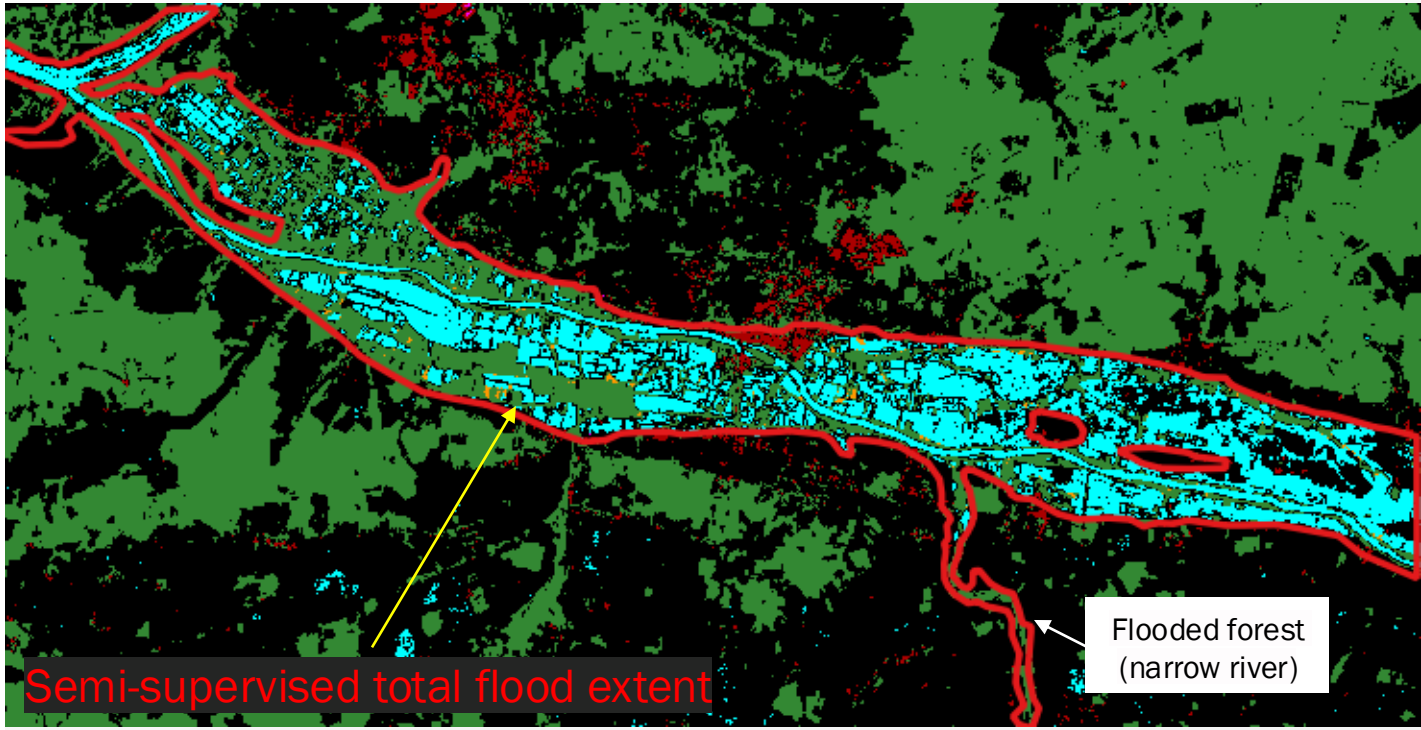


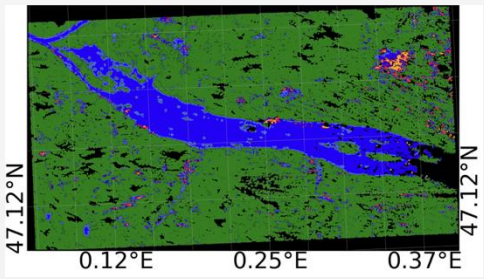


SWOT outputs on the Chinon case

Interesting new apport: **flooded forest well observed!**







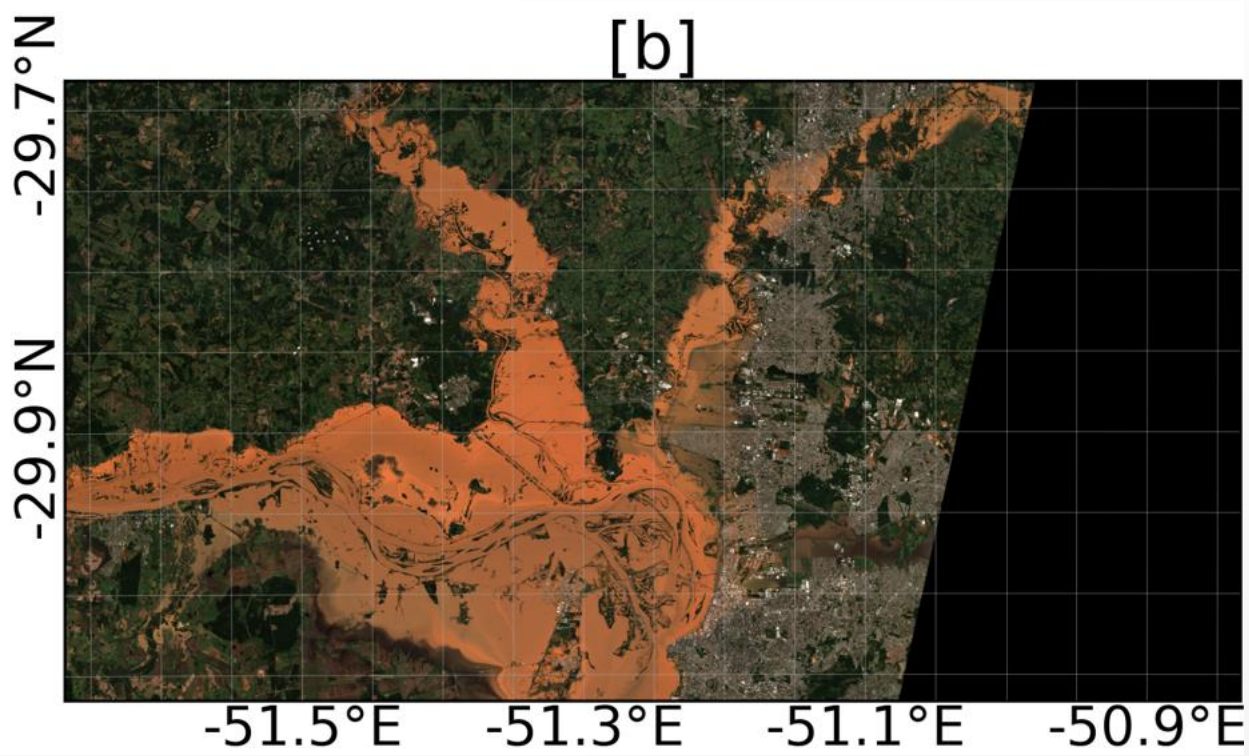
OK… and is there another more complex case?

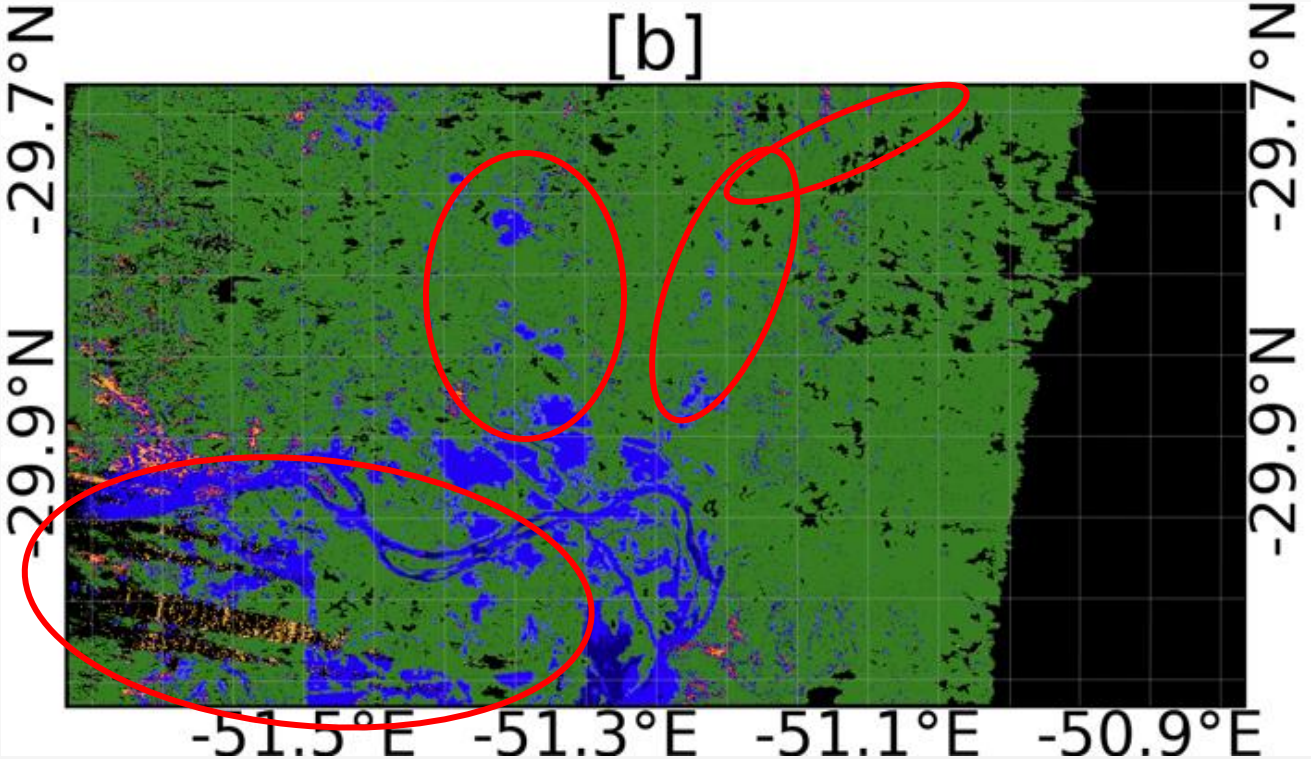
Another case: SWOT outputs on the Porto Alegre flood event

2024-05-06 – 11h46UTC

1h45 apart!

2024-05-06 – 13h31UTC



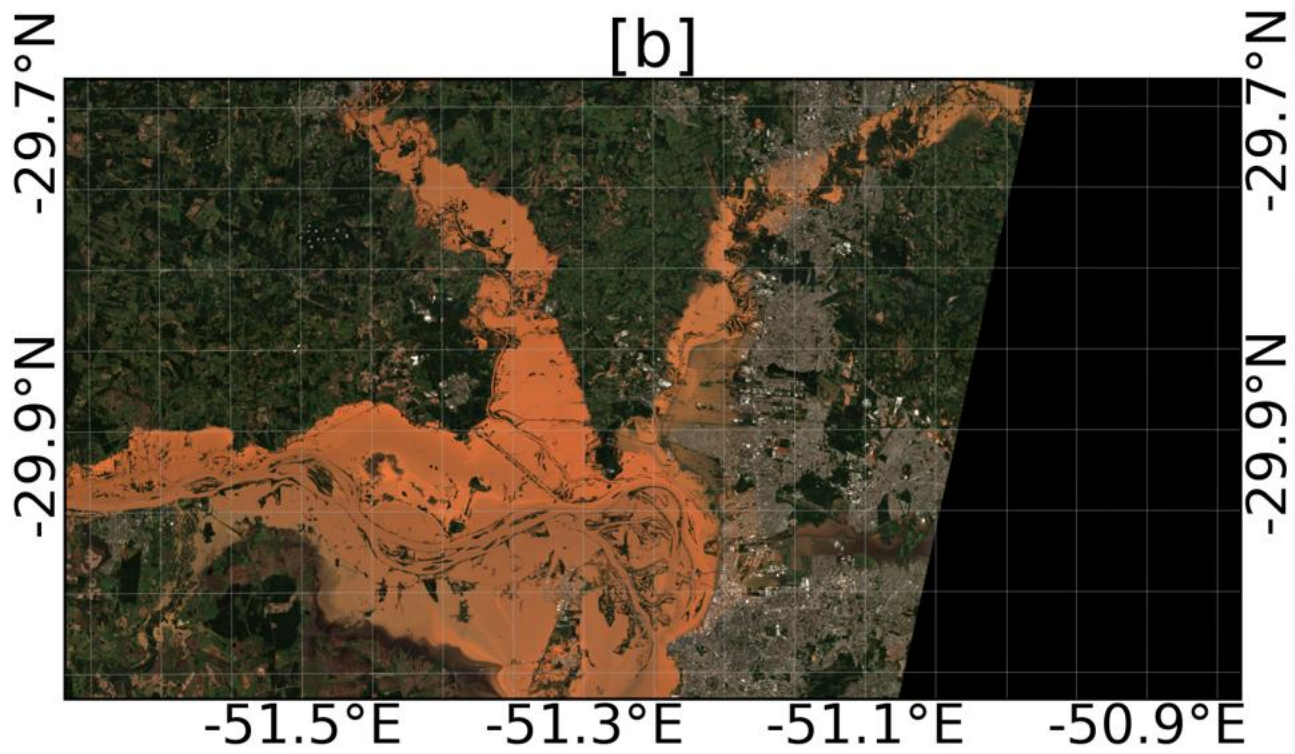


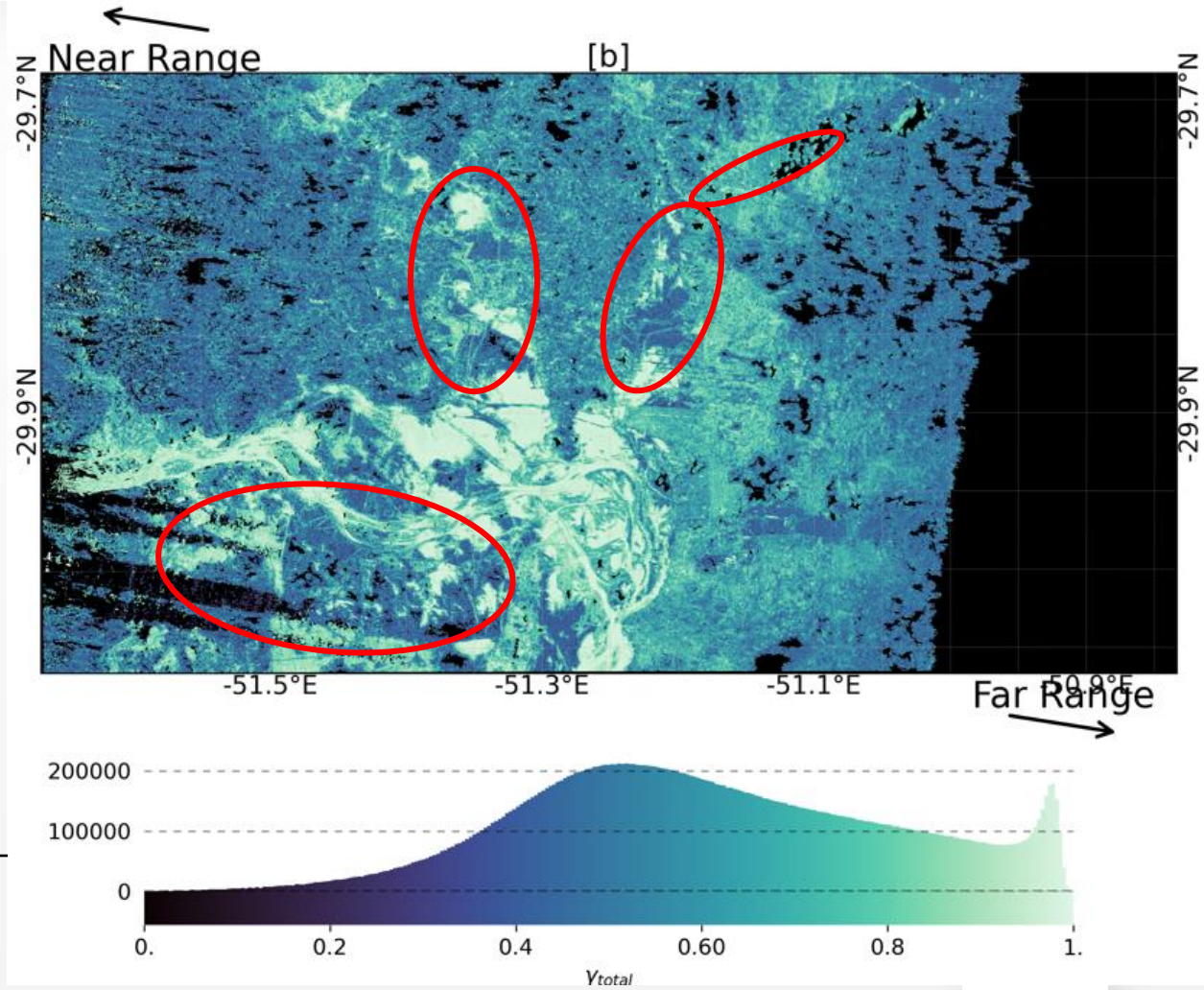
It seems the PIXC classification suffers from numerous errors

Potentially due to wind (High NE) and high-water slope?

- Land
- Land near water
- Water near land
- Open water
- Dark water
- Low-coherence water near land
- Open low-coherence water

SWOT outputs on the Porto Alegre case – Interferometric coherence

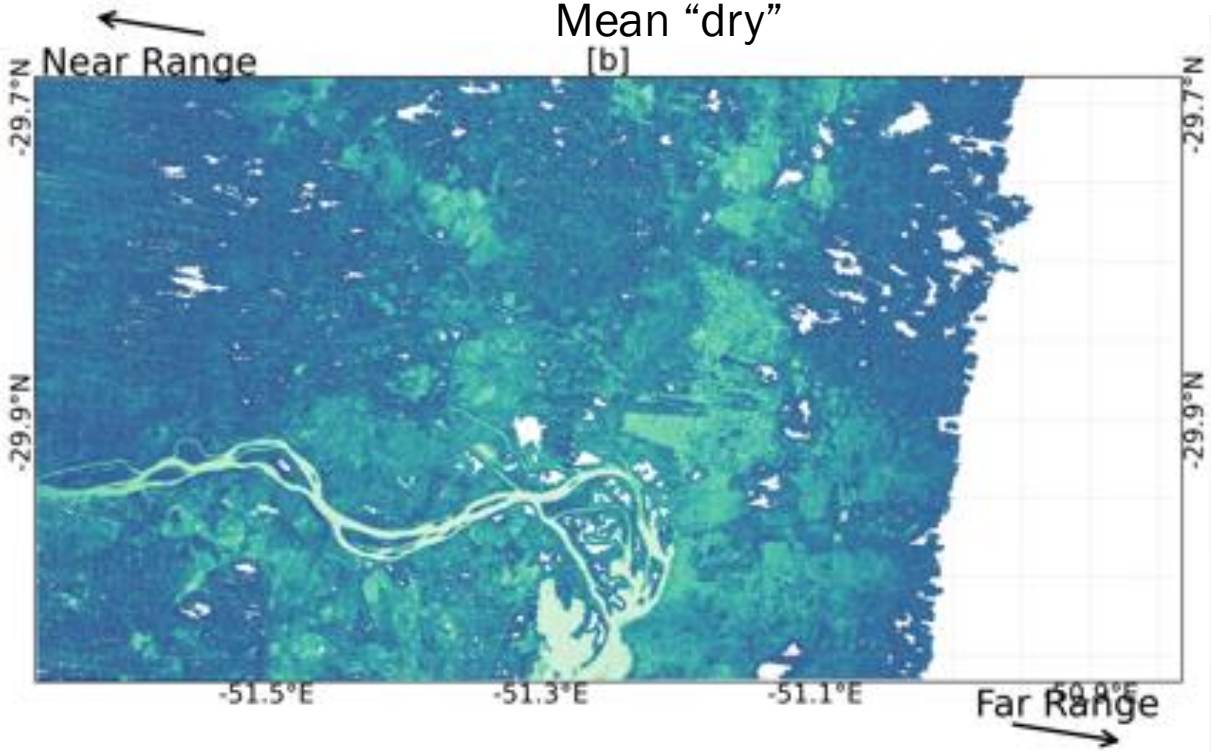




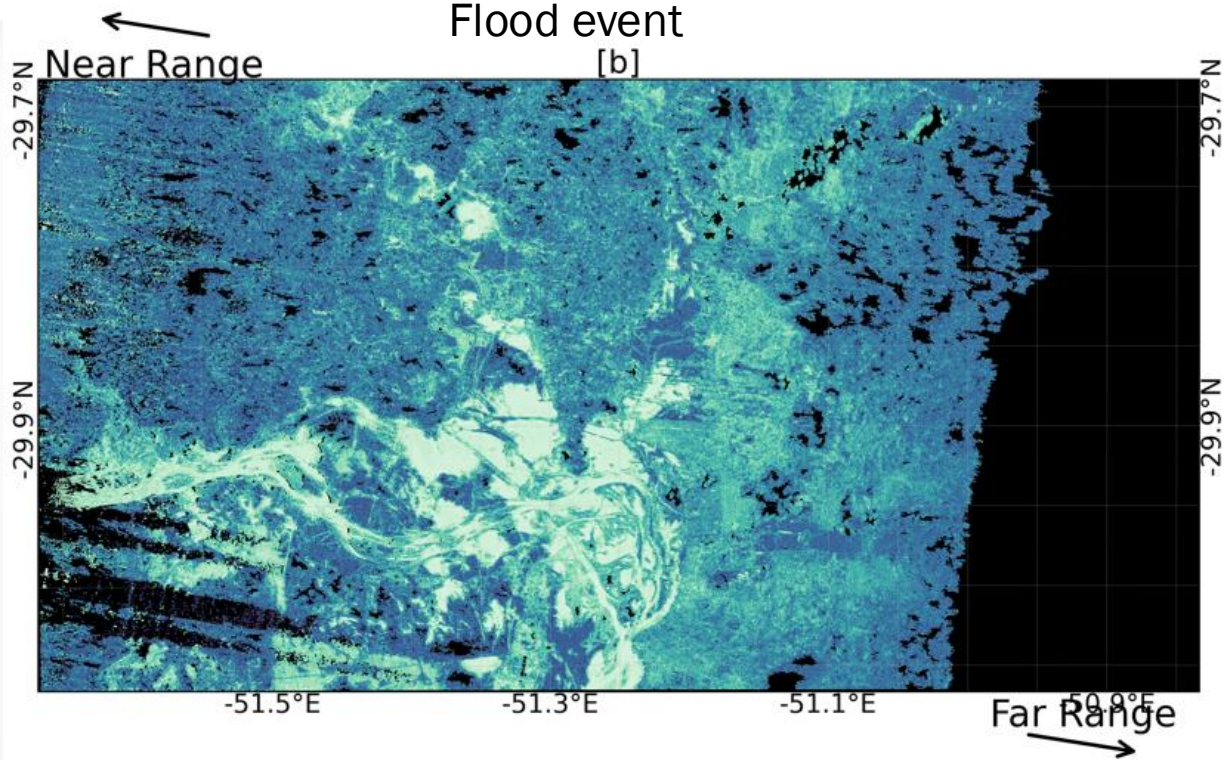
Simple thresholding interesting, but not enough…

SWOT outputs on the Porto Alegre case – Interferometric coherence

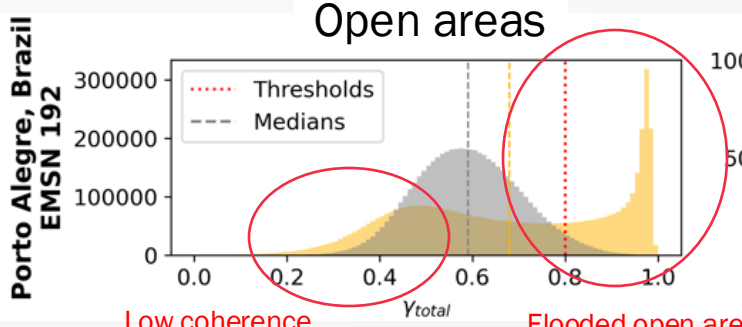
Mean “dry”



Flood event



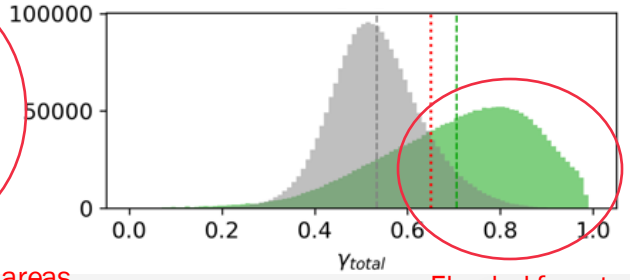
Open areas



Low coherence

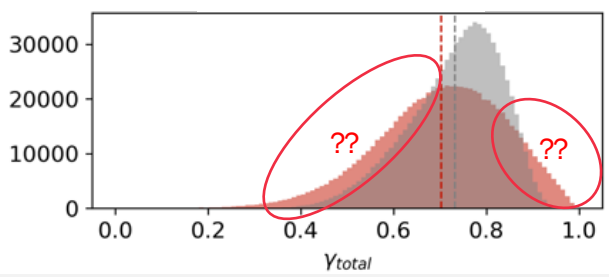
Flooded open areas

Forest areas



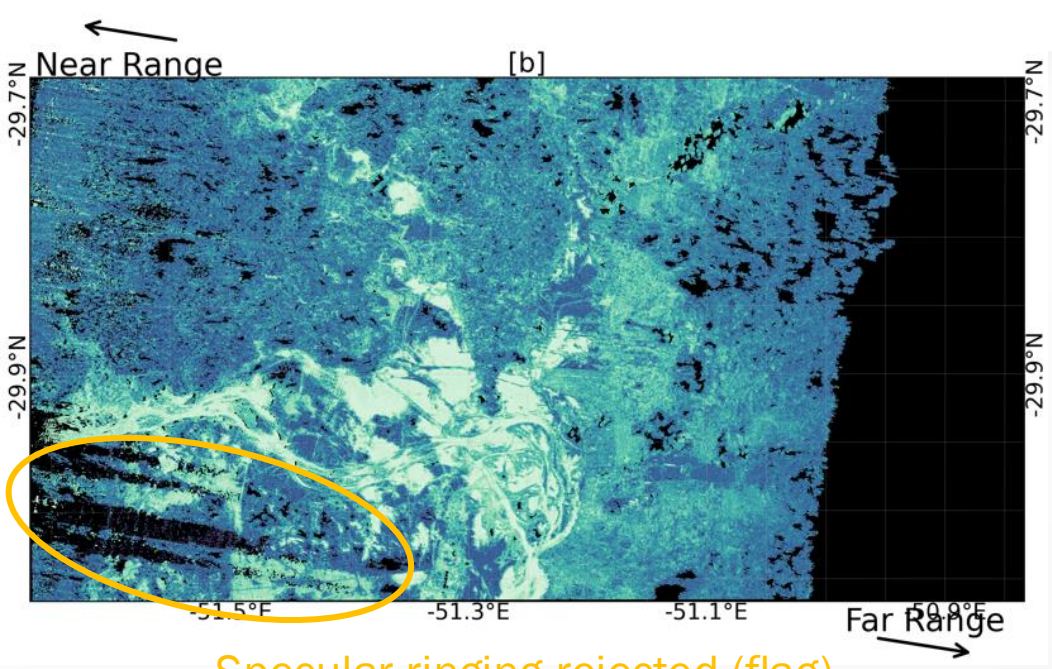
Flooded forest

Urban areas



Strange change in signal?

SWOT outputs on the Porto Alegre case – Interferometric coherence

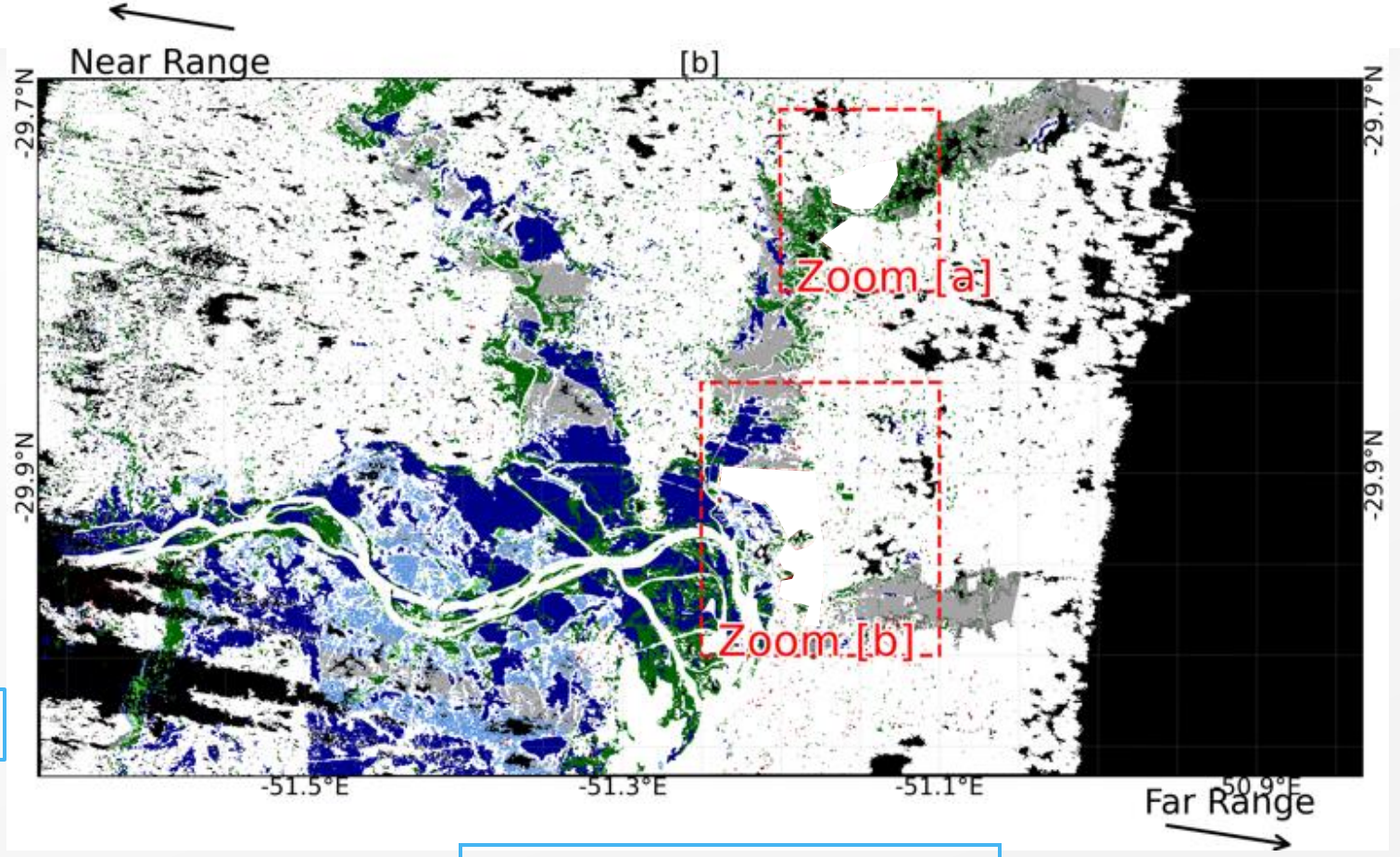


Specular ringing rejected (flag)

Low coherence + High SNR: potential dark water

Low coherence (dark water)
+ Low SNR

=> excludes unreliable areas



Flooded urban
Flooded forest
Flooded open

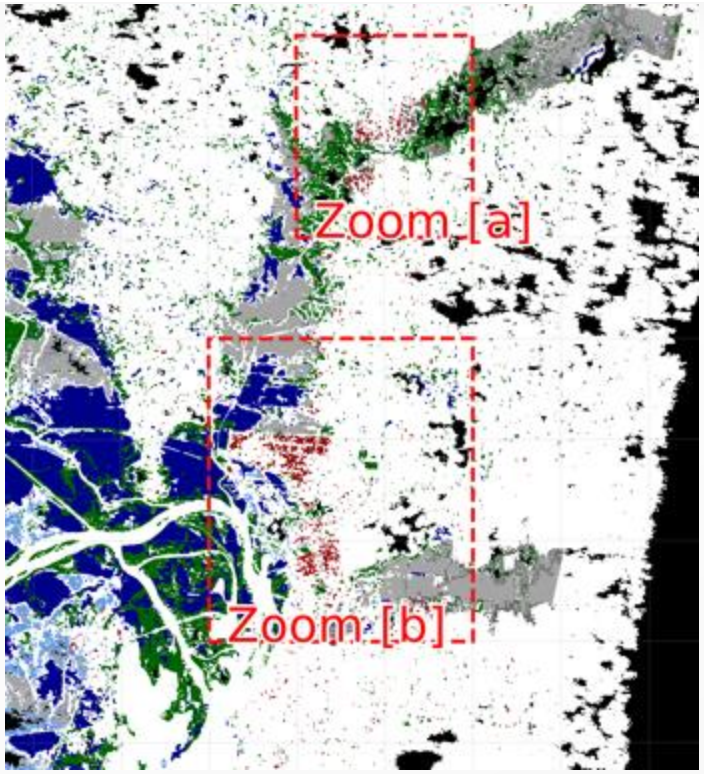
urban potential dark water
forest potential dark water
open potential dark water

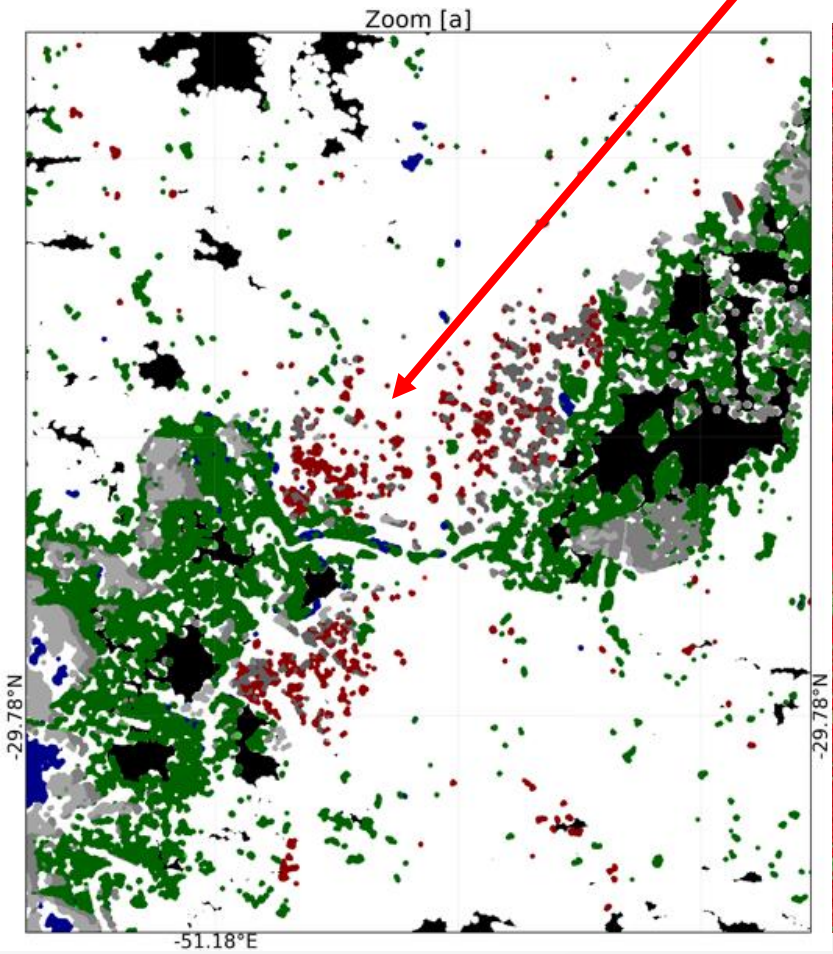
urban low SNR
forest low SNR
open low SNR

All AOI

Within Handmade flooded area

SWOT outputs on the Porto Alegre case – **Delta of interferometric coherence** (Mean/Flooded)



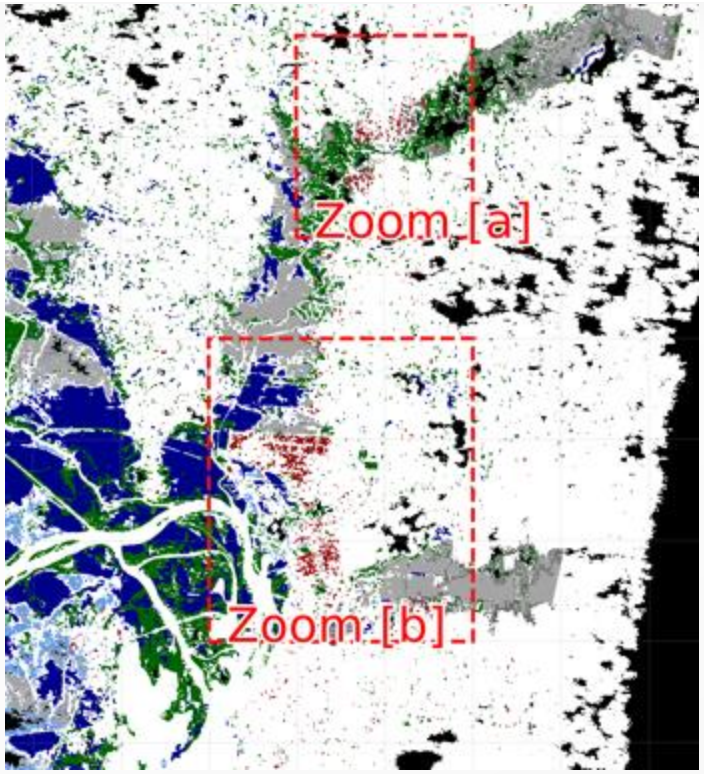


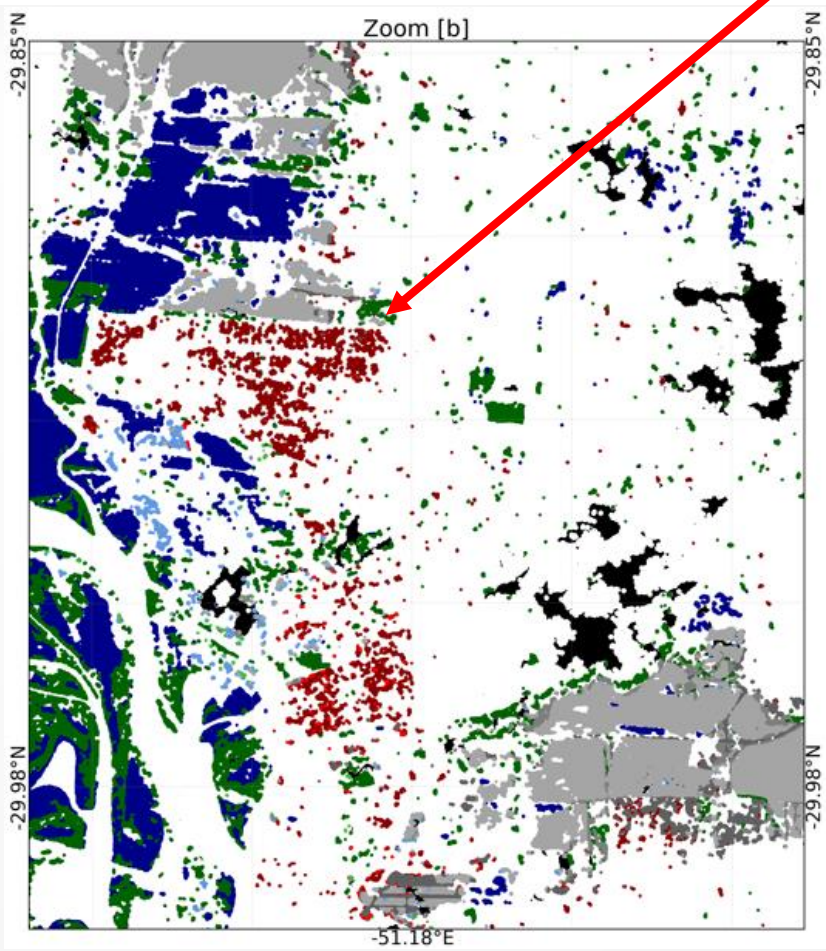


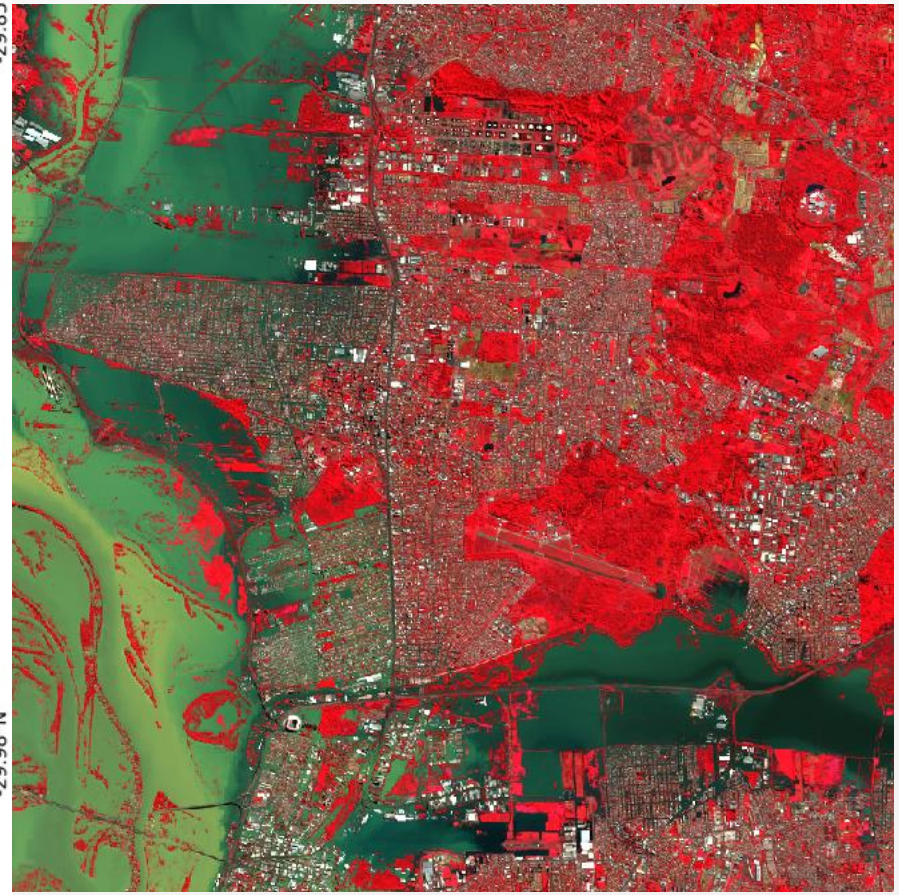
We even detect flooded urban areas directly!

Flooded urban
Flooded forest
Flooded open
urban potential dark water
forest potential dark water
open potential dark water
urban low SNR
forest low SNR
open low SNR

SWOT outputs on the Porto Alegre case – **Delta of interferometric coherence** (Mean/Flooded)







Flooded urban
Flooded forest
Flooded open
urban potential dark water
forest potential dark water
open potential dark water
urban low SNR
forest low SNR
open low SNR

We even detect flooded urban areas directly!

Take-home conclusions

New techniques to detect flooded areas from SWOT PIXC:

- Flooded forest & open areas (threshold on Sigma0, P_{coh}, γ_{tot})
- Exclusion of specular ringings with flag
- Tackle better dark water areas
  - Current Dark_water flag (within GSWE)
  - Low coherence + High SNR: potential dark water
- …but also, to detect flooded urban areas!!
  - $\Delta\gamma_{tot}$ between mean dry and flood rasters from PIXC

So many promising things with SWOT… and Sentinel-3 NG?





Thank you!!

No PhD students were harmed during this study

Bonassies et al. 2025 (2026?)

-> So close to a submission

-> Much more in depth analysis

-> Two more study cases

PhD manuscript:

In prep

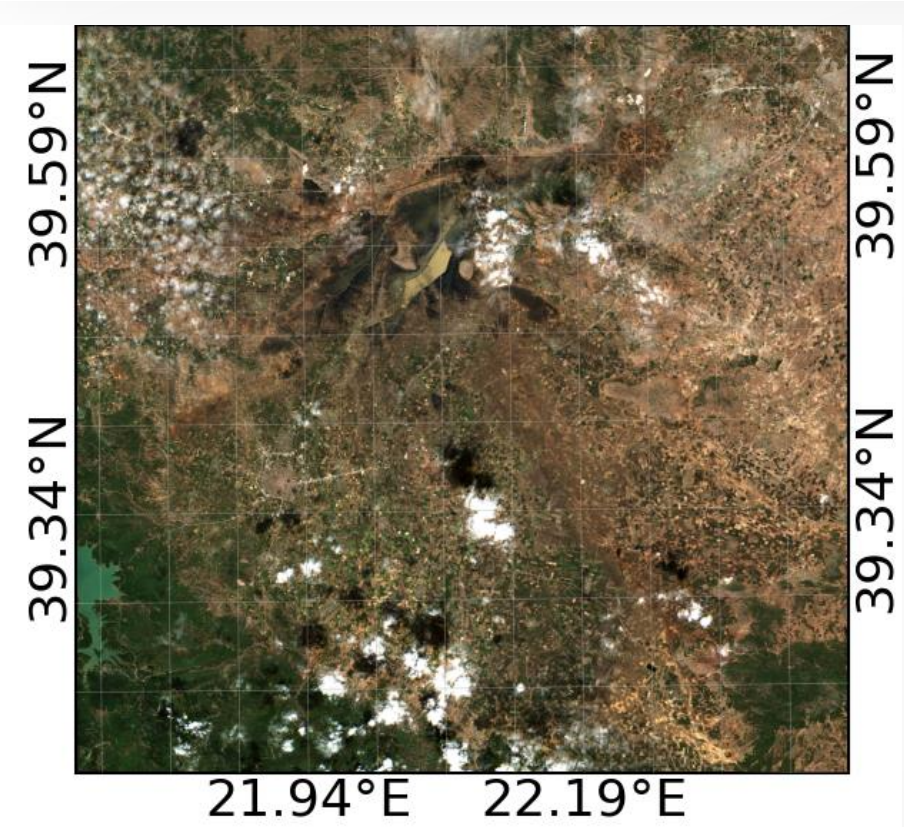
CLS COLLECTE LOCALISATION SATELLITES

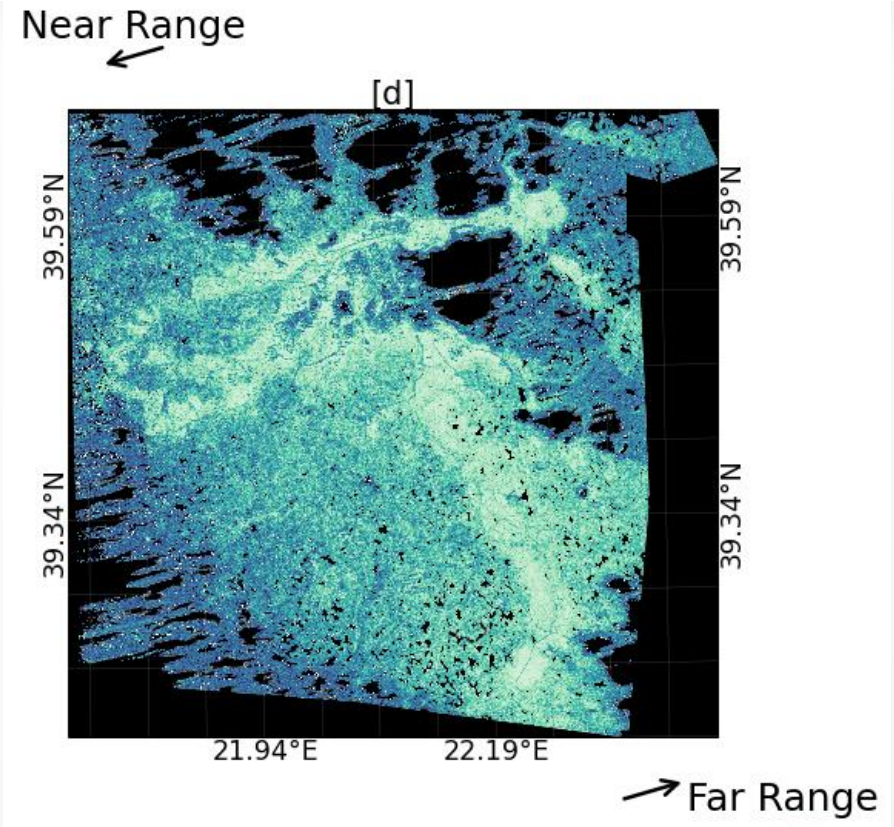
CERFACS CENTRE EUROPÉEN DE RECHERCHE ET DE FORMATION AVANCÉE EN CALCUL SCIENTIFIQUE

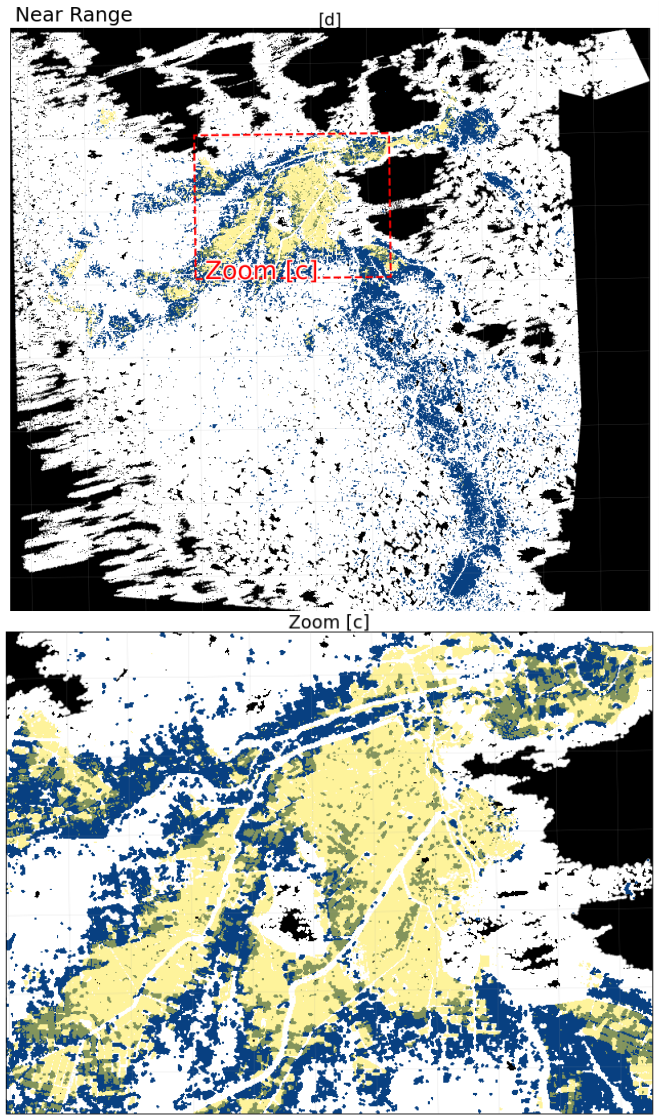
cnes CENTRE NATIONAL D'ÉTUDES SPATIALES

Contact: bonassies@cerfacs.fr

Farkadona (Greece)

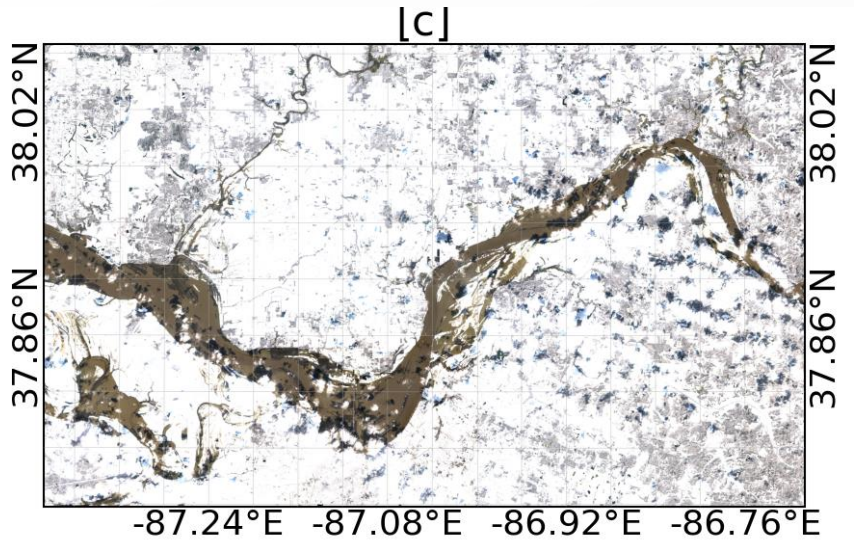


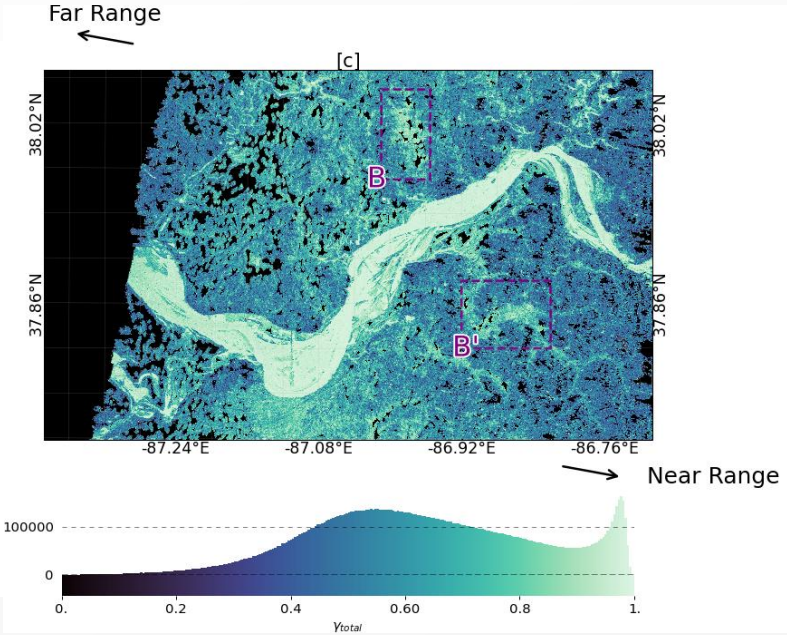


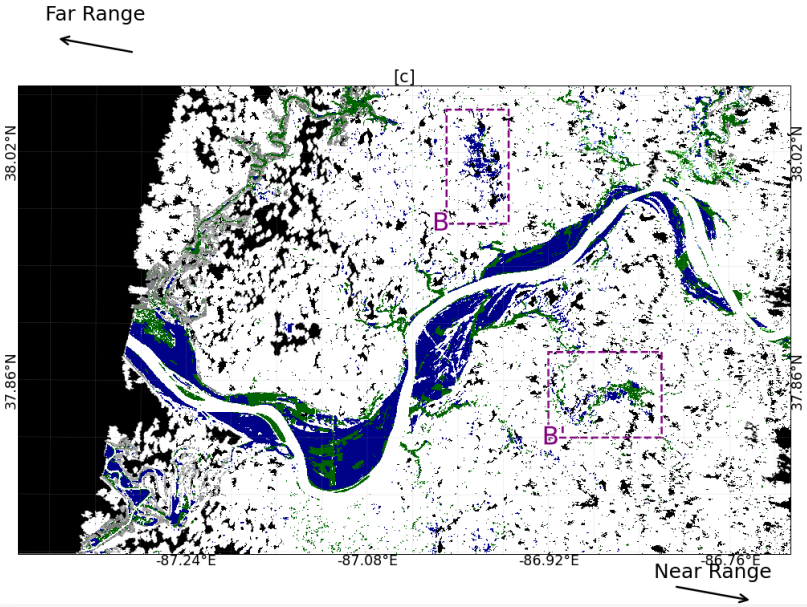


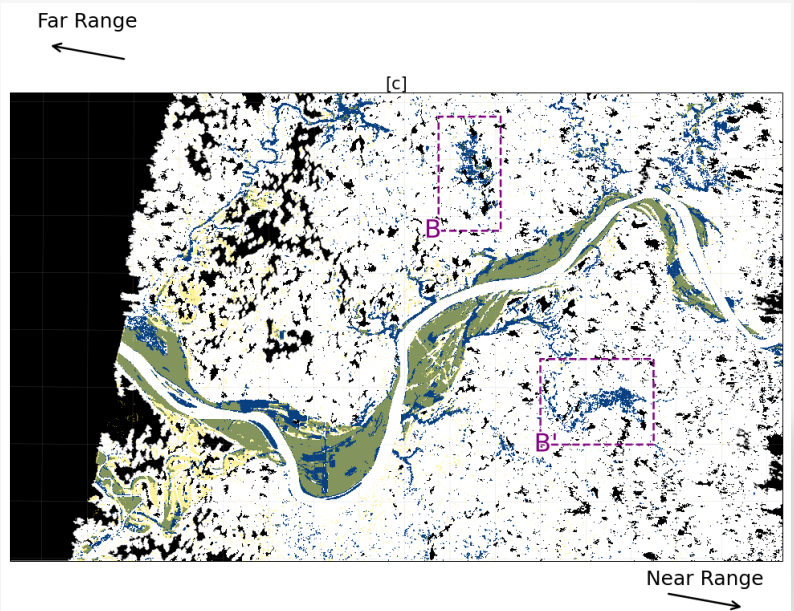
- Dark water / Low SNR
- Recessing flood -> Very high soil moisture

Owensboro (Kentucky, USA)









- Heavy snow cover
- Melting snow or…?